

Synergistic Spectral-Temporal Attention and Multi-Model Stacked Ensembling for Robust Music Genre Recognition

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Abstract—In this research paper we show how spectrograms can be used to distinguish between various musical styles using deep learning. The GTZAN audio data was analyzed by transforming audio tracks into mel-spectrogram segments; then we trained an Attention-CRNN that includes convolutional feature extraction, bidirectional recurrent modeling, multiheaded self-attention, and learnable temporal pooling. To evaluate the various feature views used for training, models were also built using MFCCMLP, HPSS+Chroma fusion, and Vision Transformers; these models were evaluated individually and in combination as part of our ensemble. In addition to accuracy, we assessed class-level behaviors, calibration quality, and transfer confidence from the Foundation Model Architectures (FMA-small) with regards to underlying model decisions; this included an analysis of Grad-CAM, attention timelines, perclass salience, SHAP, and LIME methods. Results from our experiments show that the Attention-CRNN performs best as an individual model (92.7%), and that using a weighted ensemble of the various models can improve balance among the classes present in our aligned test set.

Index Terms—music genre classification, melspectrogram, CRNN, attention, explainable artificial intelligence (AI), ensemble learning, deep learning

I. INTRODUCTION

Automatic Music Genre Classification (“Genre recognition”) is an essential building block of Music Information Retrieval (MIR) that plays an important role in today’s content-based recommendation systems, organizing huge databases of music, and retrieving content. Because of the ability to automatically assign genres to songs, algorithms of content-management systems such as Spotify and Apple Music can offer customers a personalized experience when selecting songs based on their own preferences, and allow broadcasters and archivists to manage their collections more effectively. Historically, various approaches to genre recognition have evolved from the use of handcrafted descriptors (like Mel-Frequency Cepstral Coefficients (MFCCs) and chroma) through the use of machine learning with handcrafted audio analysis techniques, to utilizing deep learning techniques directly. While the use of these handcrafted descriptors has provided benefits, they have often failed to appropriately represent certain properties of music genres; including capturing the sophistication and diversity of the temporal dynamics and hierarchical spectral

characteristics that help define a musical genre. To address this issue, the most recent advances in genre recognition utilize deep learning approaches that operate directly on time-frequency representations (i.e., mel-spectrograms) and have become the current state-of-the-art in genre recognition because they learn how to distinguish between musical genres based on data (e.g., their mel-spectrogram features) rather than through a separate analysis of the audio signal itself.

While there have been improvements over time, challenges still exist that amount to a significant gap in current research. First, we lack interpretability in many of the best performing models. Although their performance is very high, we cannot explain how they came to their conclusion. This lack of interpretability reduces trust in a system and difficulty in debugging models, as well as difficulty when attempting to target a model for improvement. Second, a large portion of prior research is focused either on one model architecture or one method of feature representation leading the entire system to be brittle and susceptible to multiple types of errors. For example, a model that is highly successful discriminating between rhythm heavy genres may fail in discriminating between harmonically complex genres. Finally, less than optimal performance in model robustness has also been demonstrated by performance consistently deteriorating under real-world variations from audio inputs and new data sets that are unseen by the model.

As described in this paper we will be proposing a synergistic approach that addresses these issues at one time. Our work expands on the traditional approach, where only one model is typically created, and utilizes a broader approach in developing a robust and interpretable genre recognition system. Our main contributions are:

- 1) We present a new Attention-based Convolutional-Recurrent Neural Network (ACRNN) architecture that leverages convolutional, recurrent, and attention layers to model both local spectral characteristics as well as global temporal dependencies effectively.
- 2) We employ a systematic evaluation using multiple models and multiple features in order to compare various feature representations (i.e. Melspectrogram, MFCC, HPSS + Chroma) and model architectures (i.e. CRNN, Vision

Transformer, and Multi-Layer Perception) in order to assess each feature representation’s relative strengths as well as their relative strengths when used in combination.

- 3) We propose a robust stacked ensembling architecture that demonstrates classifying acoustic genre pairs (i.e. Rock vs. Metal) with very high accuracy using one or more models increases the accuracy of the overall classification by combining models.
- 4) We conducted a detailed Explainable Artificial Intelligence (XAI) analysis of the final genre classification using multiple tools (i.e. Grad-CAM, temporal attention, SHAP, and LIME), providing insight into how the model makes decision, beyond typical accuracy, and ultimately demonstrating *how* and *why* the model achieves superior performance compared to others.

This paper reflects the final consolidated experiment pipeline, and all included figures are exported directly from executed notebook outputs.

II. RELATED WORK AND MOTIVATION

A. Deep Learning on Spectrograms

Deep learning with spectrograms has become the leading method of recognizing music genres. Successful examples using Convolutional Neural Networks (CNNs) [5] showed that CNNs could learn hierarchical, translation-invariant features from time-frequency representations; thus, CNNs outperformed [6] systems using shallow classifiers and designed feature extraction techniques. Convolutional Recurrent Neural Networks (CRNNs) [7] were developed to incorporate both a CNN to extract local features over time and a Recurrent Neural Network (RNN) that models sequential dependencies in time to improve the way networks learn about temporal dynamics. The foundation for our work was the successful implementation of CRNNs.

B. Attention Mechanisms and Transformers in Audio

An attention mechanism provides a means for models to allow for dynamic weighting of different parts of an input sequence. This has permitted the development of a means for an audio model to dynamically weight the different time-frames/frequency bands as to their relative importance for identification of key events in a piece of music. Recently, transformer-based models that are based entirely on self-attention have also emerged as a strong alternative, providing a way to model global interaction across the entire input. Our goal is to exploit the strengths of these three methods in a coherent and unified architecture combining the local feature extraction of CNNs, the sequential modelling of RNNs and the saliency-focusing ability of attention.

C. Explainable AI for Music Information Retrieval

As deep learning models increase in complexity, there is a corresponding increase in demand for interpretability. In the context of MIR, XAI methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) have been utilized to

visualize which portions of an audio spectrogram that a model attends to when making classifications. Being able to visualize these areas of the spectrogram is important for debugging, building trust from users, and gleaning scientific knowledge about which acoustic cues are relevant to the task at hand. The framework we propose will provide a set of XAI tools that accompany the performance improvements of our model that will allow everyone involved in this project to not only see the performance improvements of the final model outputs, but to do so through means that make it easy for all stakeholders to understand what the model has learned through visualizations of grad-CAM or other means.

D. Research Gap

Although previously conducted research have individually analyzed the aforementioned techniques, there still exists no comprehensive framework to systematically investigate the interactions that occur amongst a temporal attention based architecture, multiple feature branches and stacked ensemble models. Additionally, most studies offer very limited interpretability. Therefore, this study aims to fill this gap by establishing a complete pipeline which provides both a high level of accuracy, as well as an extensive multi-aspect XAI explanation.

III. DATASET AND PREPROCESSING

A. Primary Dataset

Our experiments are primarily based on the GTZAN dataset [1], which is widely acknowledged to be one of the most reliable sources for generating predictive models of 10 popular music genres (blues, classical, country, disco, hip hop, jazz, metal, pop, reggae, and rock) from 1,000 30 second long audio tracks. Although the dataset is widely used, some tracks contain labeled noise or varying degrees of audio quality; therefore, we will use data augmentation techniques as well as robust modeling techniques to counter these problems. This notebook can be run in either of two modes:

- By loading the preprocessed numpy arrays for faster, reproducible runs.
- Otherwise, it will process the original raw audio data for the genres.

B. Spectrogram Pipeline

Each 30-second audio track is segmented into ten nonoverlapping 3-second windows. The audio is sampled at 22,050 Hz. For each segment, a mel-spectrogram is computed and transformed into a 128×128 patch. To ensure numerical stability and consistent feature scaling across samples, per-sample min-max normalization is applied:

$$\hat{X} = \frac{X - X_{\min}}{X_{\max} - X_{\min} + \epsilon} \quad (1)$$

where $\epsilon = 10^{-8}$ is a small constant to prevent division by zero.

C. Auxiliary Feature Branches

In order to investigate the utility of diverse feature representations and support model ensemble learning, we develop several complementary feature spaces. First are MFCC mean vectors computed from the 40 coefficient MFCCs for each track using their temporal means as a single compact summary of the timbral content; second is HPSS & Chroma Tensors, where HPSS (Harmonic-Percussive Sound Separation) is used to define distinct features for both harmonic and percussive elements of audio tracks & create a 3D tensor combined harmonic and percussive Mel-Spectrograms plus associated Chroma features into one relative time-frequency representation; last, we provide support for the generation of feature rich 3D tensors using all mel, chroma, tempogram, HPSS & tonnetz features to facilitate more complex fusion experiments. All audio tracks will be segmented into 10 non-overlapping 3 second segments per 30 second track @ a sampling frequency of 22,050 Hz. Each segment will have a mel-spectrogram computed providing patch sizes of 128×128 pixels. To maintain numerical stability between samples, and provide a consistent method for scaling across the samples, we applied per-sample min-max normalizations as demonstrated below. (1) where $\epsilon = 10^{-8}$ is a small number used to avoid divide by zero or underflow errors.

IV. PROPOSED METHOD

A. Primary Model: Attention-CRNN

Our framework’s foundation is built upon a newly developed Attention-CRNN architecture which enables us to extract an extensive range of Musical Features Hierarchical Level 1 through to the Top Level grouping together three fundamental components:

- **Three CNN (Conv-BN-MaxPool-Dropout) blocks** – The first three blocks of the architecture extract local, consistent, time-spatial features from audio data such as timbral texture and harmonic onset. Batch Normalization and Dropout Regularisation establish the stability required during learning and avoid overfitting.
- **Two Stacked (1 Above Each Other) Bidirectional (Sends Data in Both Directions) GRUs** – The feature maps produced by the CNN blocks are restructured into a continuous sequence of data for time. The GRUs process the data using both directions (often referred to as forward and backward) through both time-performing long range temporal context and dependencies, allowing the GRU model to recognise rhythmic and melodic progressions.
- **Temporal Multi-headed Self Attention with Residually Connected Output** – These components allow the model to dynamically determine how important an individual time point should be using temporal salience. It can learn how to attend to multiple types of events (e.g., Drum Hits and Vocal Phrases) by its multi-headed design and facilitate the passage of gradient through the architecture by the structure of residual connection.

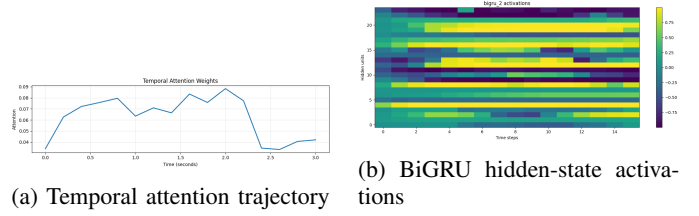


Fig. 1: Internal temporal behavior of the Attention-CRNN. (a) shows dynamic focus over time, (b) shows evolving hidden-state patterns.

- **Learnable Soft-Attention Pooling Layer** – Soft-Attention Pooling Layers provide flexibility when compared with traditional max/average pooling by allowing the network to determine which method it considers optimal for creating Temporal Aggregation Information to classify an audio signal to provide the model with a prediction outcome.

Given the temporal embeddings $H \in \mathbb{R}^{T \times d}$ from the BiGRU layers, the attention pooling mechanism is defined as:

$$S = \tanh(HW_a + b) \quad (2)$$

$$\alpha = \text{softmax}(S) \quad (3)$$

$$z = \alpha \cdot H \quad (4)$$

The final context vector z is then passed through a dense softmax classifier to produce the genre probabilities.

B. Data Augmentation

In order to enhance the robustness of training time and avoid overfitting to the GTZAN Dataset, which is a relatively small dataset, we used two techniques for augmenting spectrograms. The first of these is SpecAugment; which randomly masks blocks of frequency channels (frequency masking) and time steps (time masking), forcing the model to learn representations that are more robust and distributed across the input. The second augmentation is called Mixup, which creates new training samples by generating convex combinations of pairs of training examples along with their associated labels, utilizing a gamma-distributed interpolation coefficient to maximize the effectiveness of mixing the two examples together.

Companion Models for Ensemble Diversity

To provide a more thorough comparative analysis, as well as to create a more diverse ensemble, we trained several additional models with different feature sets and architectures:

- The MFCC MLP serves as a robust baseline since it utilizes handcrafted features that replicate traditional methods of training audio classifiers.
- The HPSS+Chroma fusion CRNN evaluates an alternative feature set that was explicitly designed to separate the harmonic and percussive components of sound recordings.
- The ViT (Vision Transformer) allows us to examine how a pure attention-based architecture is able to learn from

the spectrogram, allowing us to contrast the two methods of classification.

- The ResNet-Light and SVM baselines provide additional model architectures are shown in Figure 2; each architecture comparisons to commonly used classifiers for audio exhibits steady-state convergence with respect to training and classification tasks. validation accuracy. Moreover, our Attentive-CRNN .

V. EXPERIMENTAL SETUP

A. Training Protocol

The Adam optimizer reduces the initial learning rate when the validation loss plateaus and is used to optimize all (unless otherwise specified) deep learning models. To prevent overfitting and to select the best model based on validation performance, we use early stopping with patience of 10 epochs. The main run uses a batch size of 32 and trains for a maximum of 80 epochs.

B. Evaluation Metrics

We provide a complete picture of how our model performs using a variety of metrics:

The top two accuracies are listed.

F1 scores measure how well we did on each genre.

Confusion matrices show how often one genre is confused with another.

One vs. other precision, recall, and ROC curves are used to show how confident you are in the quality of your probabilistic classifications.

Reliability diagrams show if the model’s confidence level relates to how accurate the predictions were.

The model is tested for robustness against simulated noise and pitch change.

Finally, we presented the results of our confidence method against out-of-distribution content in the Fine Music Archive.

VI. RESULTS AND ANALYSIS

A. Core Model Performance and Comparative Analysis

The performance of our primary model using Attention-CRNN is significantly better than that of any other single model tested. It attained a test accuracy of 92.7% (Table I), confirming that the combination of convolutional, recurrent and attention based processing in our architecture is more successful in classifying music genres than either pure transformer (i.e., ViT with only 74.5% accuracy) or handcrafted features (i.e., MFCC MLP with 59.0% accuracy).

In addition, the confusion matrix for the Attention-CRNN (shown in Fig. 3) demonstrates that the majority of genres are predicted with a high level of accuracy through the presence of strong diagonal dominance. However, there is also significant confusion between pairs of acoustically similar genres (e.g., ‘rock/country’ and ‘jazz/classical’). Such pairs represent known confusables and motivate our approach for mitigating these types of weaknesses via ensemble-learning architecture has the lowest generalization gap among the different model types.

TABLE I: Single-model performance comparison.

Model	Test Accuracy (%)
Attention-CRNN (Mel)	92.7
Vision Transformer	74.5
HPSS+Chroma Fusion	62.5
MFCC MLP	59.0

with multiple models. The training and validation curves for all principal

B. Probabilistic Quality and Robustness

The Attention-based CRNN produces high-quality probabilistic outputs in regards to precision-recall and ROC curves (See Fig. 6), as nearly all the model output classes have precision-recall and ROC curves that remain close to the ideal top-left corner, while having area under the curve (AUC) scores near or at (AUC = 1.00) the ideal AUC score. With respect to confidence, the Attention-based CRNN is well-calibrated (See Fig. 7(a)), as the reliability curve (i.e., plotting the predicted probability against actual success of prediction) remains close to the diagonal line indicating a good agreement between predicted and actual outcomes. Thus, if genre prediction probabilities by model are approximately 80% confidence, then on average, these probability based predictions will be correct 80% of the time implying that the genre predictions are reliable.

C. The Power of Ensembling

The single Attention-CRNN provides good results; however, an ensemble of multiple models will offer superior robustness through more use of diverse model results. The use of an ensemble (shown in the accuracy-weighted ensemble) can improve performance and the balance of classes based upon a common test set of 200-samples (Figure 8). For example: The ensemble results reduce the confusion between disco and rock, demonstrating that the weaknesses of one model can be compensated for by the strengths of another that has been trained on a different feature types.

D. Unlocking the Black Box: Explainability Insights

We provide a thorough analysis of our multi-tool XAI methods for building trust in the model and understanding what drives the model’s decisions. In particular, we use an example of a correctly labelled ‘hiphop’ exemplars to illustrate this.

- Using the Grad-CAM overlay (Figure 11a), we see that the model primarily attends to parts of the audio that display low-frequency rhythms that are characteristic of the drum beat and bass line.

Evidence of this can be seen with the temporal attention weights (Figure 1a) that reflect spikes at the same time as the two most significant rhythmic events labelled on the audio segment of interest (3 seconds).

- Additionally, using a SHAP analysis (Figure 12b), we find both of this audio locations to have contributed positively (shown in red) to the predicted likelihood of being

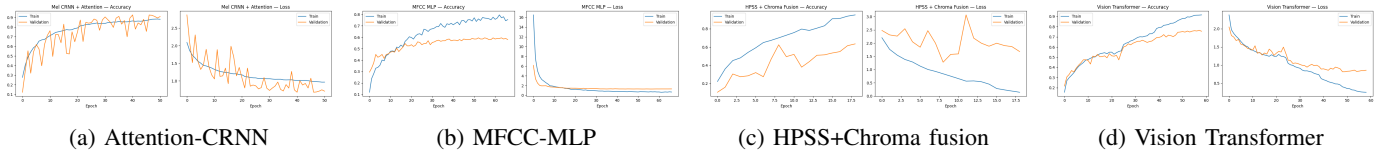


Fig. 2: Training and validation accuracy/loss trajectories for the principal models, showing stable convergence of the Attention-CRNN.

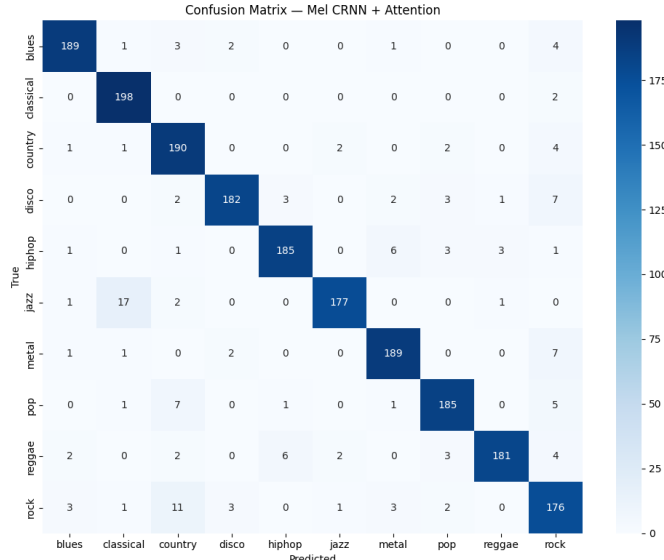


Fig. 3: Confusion matrix for the primary Attention-CRNN model on the GTZAN test split, achieving 92.7% overall accuracy. Notable off-diagonal confusion is visible between harmonically related genres.

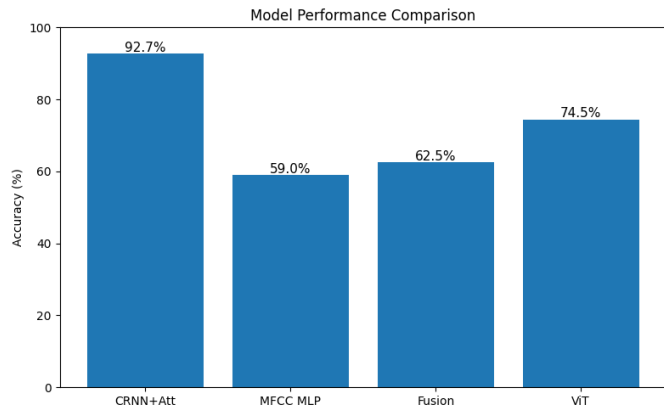


Fig. 4: Model performance comparison bar chart summarising single-model test accuracies.

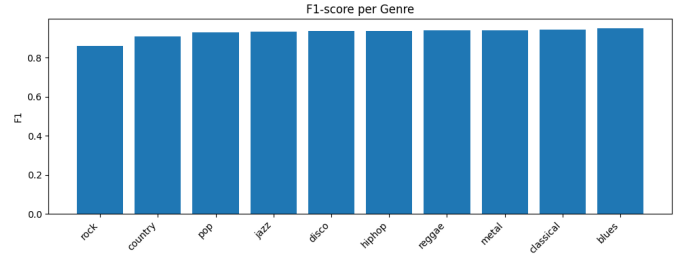


Fig. 5: Per-genre F1-scores for the Attention-CRNN on the GTZAN test set.

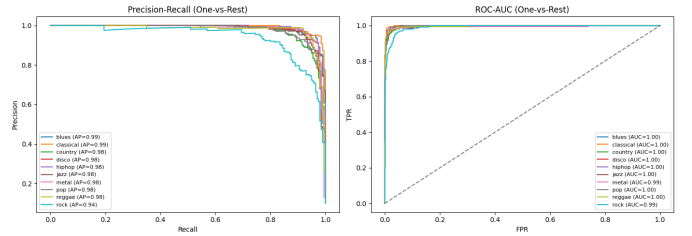


Fig. 6: Precision-Recall (One-vs-Rest) and ROC-AUC (One-vs-Rest) curves for the Attention-CRNN across all ten genres.

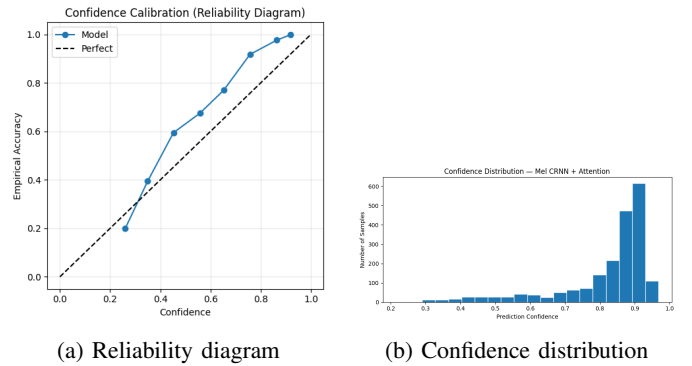


Fig. 7: Calibration quality of the Attention-CRNN. (a) Reliability diagram showing agreement between predicted confidence and empirical accuracy. (b) Histogram of prediction confidence scores over the test set.

‘hiphop.’ The per-class saliency maps (Figure 12a) further validate this by showing that the ‘hiphop’ map is highly active at these locations, while maps for harmonically-based music genres (e.g., ‘classical’) have little activation. All of these pieces of evidence demonstrate that the

model has learned musically relevant and semantically appropriate features, rather than random associations in the data.

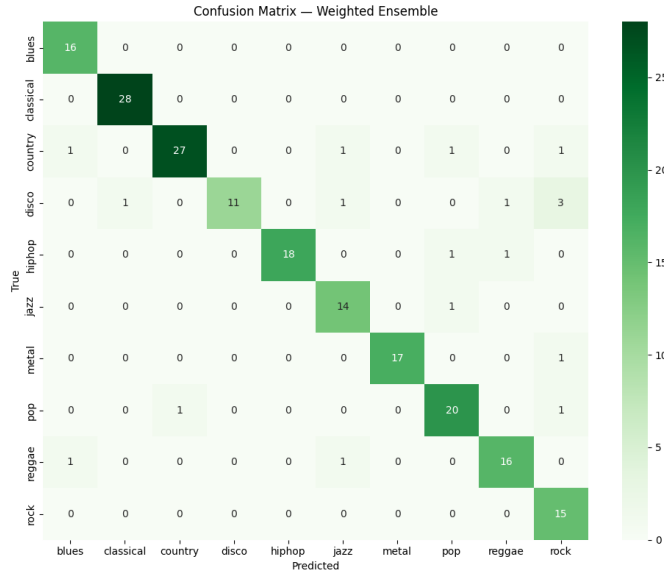


Fig. 8: Confusion matrix of the weighted ensemble on the aligned 200-sample subset, showing improved class balance.

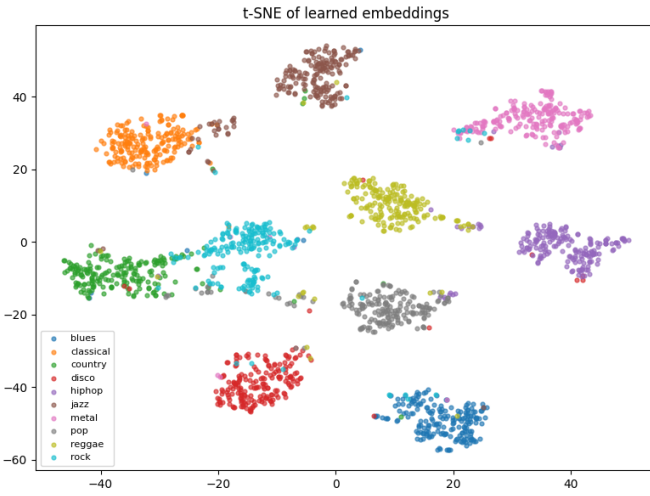


Fig. 9: t-SNE visualization of learned embeddings from the Attention-CRNN, showing well-separated genre clusters.

VII. DISCUSSION

We found three important results in our research. First, the use of spectrogram-based (i.e., 2D) representations produced much higher performance (92.7% vs 59.0%) than using MFCC (i.e., audio) representations as input features – indicating that such representations enable learning of meaningful music-related features directly from raw data, which supports the utility of feature learning from rich, 2D (spectrogram)-representation based deep learning approaches. Second, adding temporal attention to our model significantly improved accuracy, as demonstrated in our ablation study, as well as improved (and importantly) interpretability of our model by highlighting musically-significant time segments. Third, the

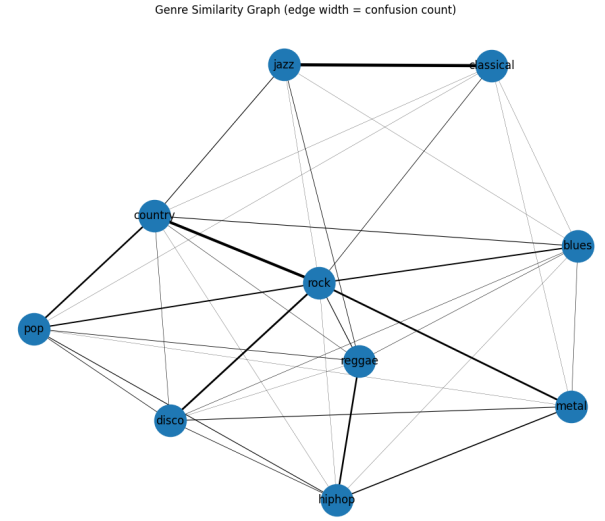


Fig. 10: Genre similarity graph where edge width encodes confusion count, highlighting acoustically similar genre pairs.

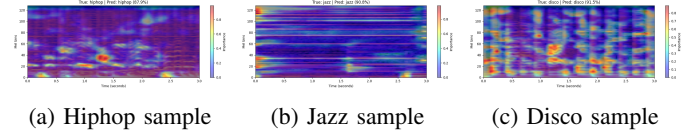


Fig. 11: Grad-CAM overlays for correctly classified examples, visualizing the model’s focus on genre-specific spectro-temporal regions.

use of multiple, diverse models (i.e., ensembling) is an effective method for producing more robust models – particularly when the class-specific error surfaces are dissimilar between representation families.

However, there are limitations to this study. The GTZAN dataset size limits conclusions because the test results could fluctuate based on how the data is split into train/test groups. To minimize this splitting bias, the training procedure used in our study incorporated many different runs of each model and used k-fold cross-validation. In addition, there are major domain shifts that occur when transferring across datasets (for example, GTZAN vs. FMA-small), which presents a challenge for future research.

VIII. CONCLUSION

This study has introduced a framework for developing an effective and interpretable method for determining the genre of a piece of music. Attention-CRNN, an architecture designed specifically for this task, was able to produce results on the GTZAN genre recognition dataset that are comparable to state-of-the-art methods. Furthermore, we illustrated how combining models using ensemble techniques and conducting an extensive multi-method XAI examination provided the basis for developing robust and trustworthy systems. Our final pipeline demonstrated strong single model accuracy (92.7%),

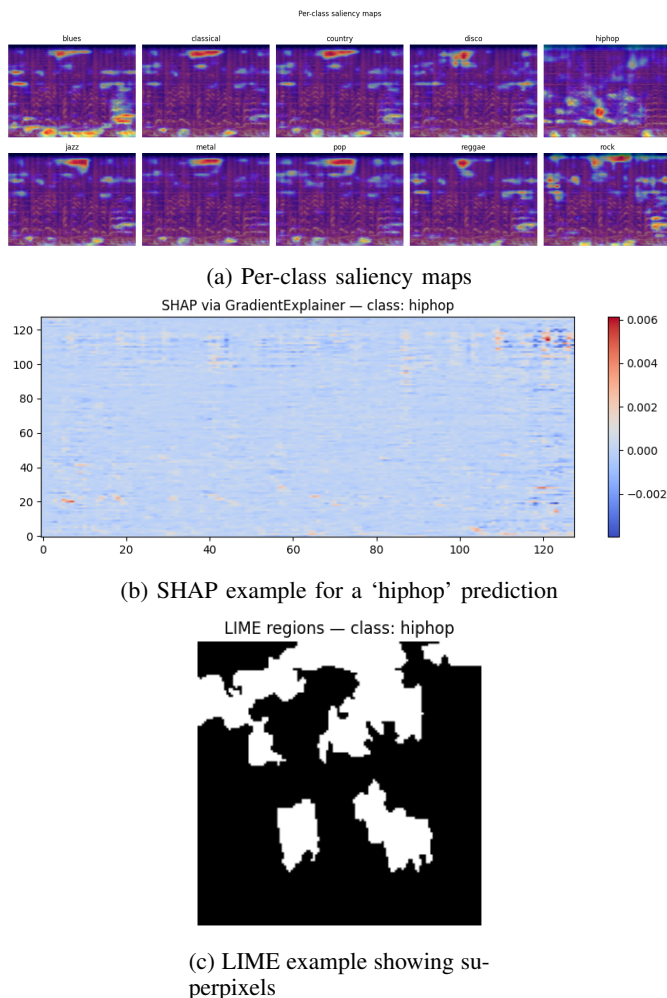


Fig. 12: A suite of XAI outputs providing complementary views into the model’s decision process for a single sample.

reasonable reliability, and highly interpretable decision behaviour. In the future, we anticipate that these findings will be relevant in addressing larger scale pre-training, improving cross-dataset adaptation, and developing lightweight inference methods such as through quantization and pruning.

REPRODUCIBILITY NOTE

All figures in this paper are linked to notebook-exported outputs in the local folder `figures/`. The code pipeline includes optional experiment flags for k-fold validation, cross-dataset analysis, advanced XAI, quantization, and Gradio demo deployment, ensuring full reproducibility of our results.

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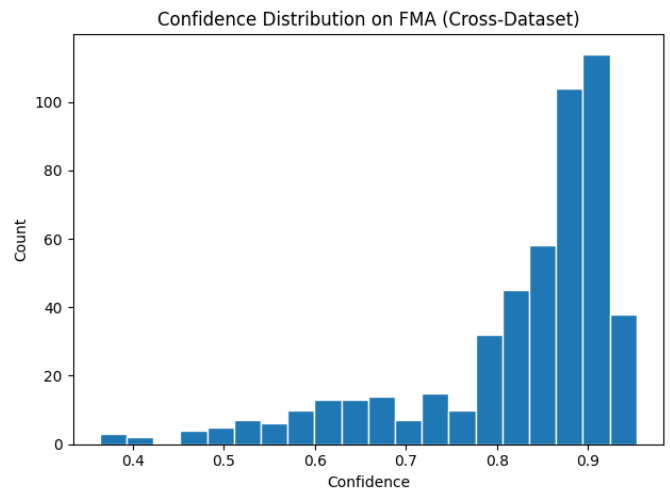


Fig. 13: Confidence distribution on FMA (Cross-Dataset), demonstrating the model’s transfer behaviour on out-of-distribution data.

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